UNIT – 1

ASSESSMENT – 2

REPORT

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1. Title

Doctor Shift Scheduling using Backtracking Search Algorithm

Objective

To assign doctors to hospital shifts while satisfying all scheduling constraints using the **Backtracking Search Algorithm**.

Theory

Scheduling problems are examples of **Constraint Satisfaction Problems (CSPs)**.

In a CSP, variables must be assigned values while satisfying a set of constraints.

The **Backtracking Search Algorithm** explores possible assignments recursively. If a constraint is violated, the algorithm returns to the previous assignment and tries another value.

This method avoids checking every possible combination and efficiently finds a valid solution.

Algorithm

1. Start with no assignments.
2. Select the first doctor.

3. Assign a valid shift.
4. Check all constraints.
5. If constraints are satisfied, continue to the next doctor.
6. If any constraint fails, backtrack and try another shift.
7. Repeat until all doctors are assigned.
8. Display the valid schedule.

Advantages

- Efficient for constraint satisfaction problems.
- Avoids invalid assignments early.
- Guarantees finding a valid solution if one exists.
- Easy to implement for scheduling problems.

Applications

- Hospital duty scheduling
- Employee shift allocation
- Examination timetables
- Resource scheduling
- AI planning systems

Result

The **Backtracking Search Algorithm** successfully assigned the doctors to shifts while satisfying all the given constraints.

Final Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

All constraints were successfully satisfied, demonstrating the effectiveness of the Backtracking Search Algorithm for solving constraint satisfaction problems.

2. Title

Robot Path Planning using Breadth-First Search (BFS)

Objective

To determine the shortest path for a robot in a 5 × 5 grid using the Breadth-First Search algorithm while avoiding obstacles.

Theory

Breadth-First Search (BFS) is a graph traversal algorithm that explores nodes level by level. Since every movement in the grid has the same cost (1), BFS always finds the shortest path from the start position to the goal.

The Manhattan Distance heuristic estimates the distance between the current position and the goal. Although BFS does not use heuristics for decision-making, Manhattan Distance helps estimate the remaining distance in grid-based navigation.

Algorithm

1. **Start at the initial node S.**
2. **Add the start node to the queue.**
3. **Remove the front node from the queue.**
4. **Expand all valid neighboring nodes.**
5. **Ignore obstacles and already visited nodes.**
6. **Add new nodes to the queue.**
7. **Repeat until the goal G is reached.**
8. **Display the shortest path and total cost.**

Advantages

- **Guarantees the shortest path when all move costs are equal.**
- **Simple and easy to implement.**
- **Complete algorithm (finds a solution if one exists).**
- **Suitable for grid-based pathfinding.**

Applications

- **Robot navigation**
- **Game AI**
- **Maze solving**
- **Network routing**
- **Path planning**

Result

The Breadth-First Search (BFS) algorithm successfully determined the shortest path from the Start (S) to the Goal (G) while avoiding obstacles. The robot reached the destination with an optimal path cost of 8, demonstrating the effectiveness of BFS for uniform-cost grid navigation.

3. Title

Autonomous Rescue Robot using Online Search and Uniform Cost Search

Objective

To develop an AI-based solution for an autonomous rescue robot operating in a disaster-affected building using an Online Search Agent and Uniform Cost Search.

Theory

Autonomous rescue robots are widely used in disaster management to search for survivors in dangerous environments.

Since the robot has only partial knowledge of its surroundings, it must continuously gather information, avoid obstacles, and update its path dynamically.

The Online Search Agent allows the robot to make decisions based on newly observed information.

Uniform Cost Search ensures that the chosen path has the lowest cumulative cost while considering risky zones.

Algorithm

1. Start from the initial position **S**.
2. Observe neighboring cells.
3. Ignore blocked cells.
4. Calculate movement cost.
5. Add extra cost for risky zones.
6. Choose the least-cost path using Uniform Cost Search.
7. Update knowledge after every move.
8. Repeat until the survivor **G** is reached.

AI Concepts Used

- Intelligent Agent
- Rational Decision Making
- Online Search
- Uniform Cost Search
- Partially Observable Environment
- Dynamic Environment

Applications

- Disaster Management
- Search and Rescue Missions

- Military Robotics
- Space Exploration
- Warehouse Automation
- Autonomous Navigation

Advantages

- Adapts to unknown environments.
- Minimizes travel cost.
- Avoids dangerous paths.
- Efficient path planning.
- Suitable for real-time rescue operations.

Result

The autonomous rescue robot successfully demonstrated intelligent decision-making using an **Online Search Agent** and **Uniform Cost Search (UCS)**. The robot dynamically updated its knowledge, avoided obstacles, minimized the total travel cost, and safely reached the survivor, making this approach well-suited for disaster response scenarios.